DATABASE ARCHITECTURE SPECIFICATION

PC CRM Enterprise Platform

PC CRM Database Architecture & Storage Engine Manual

A comprehensive deep-dive into the relational schema, storage engine mechanics, table relationships, metadata catalog, and MariaDB optimization of PC CRM.

MariaDB 11.x Enterprise | InnoDB ACID Transactions | Entity-Relationship Layout | Child Table Parentage | Single-Row Settings Pattern | Audit Trail & Versioning | Indexing & Query Optimization

PC CRM Database Engineering • Version 2026.1

Target Audience: System Architects, Database Administrators & Technical Leads

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1. Why MariaDB for PC CRM?

The foundation of PC CRM is built upon **MariaDB**. The architectural decision to standardize on MariaDB over other database engines is rooted in four essential enterprise requirements: transactional reliability, high-concurrency performance, native JSON flexibility, and open enterprise governance.

1. STRICT ACID COMPLIANCE WITH INNODB

In customer relationship and sales management, monetary transactions, deal stages, and customer conversions must never experience dirty reads or lost updates. MariaDB delivers true ACID guarantees via the InnoDB storage engine, with row-level locking and multi-version concurrency control (MVCC).

2. HYBRID RELATIONAL + DOCUMENT FLEXIBILITY

PC CRM combines strict relational tables (Leads, Deals, Organizations) with dynamic JSON fields for flexible payloads (e.g., custom form layouts, webhook responses, geolocation tracking). MariaDB native JSON functions allow instant indexing and querying of nested JSON attributes without sacrificing relational integrity.

3. HIGH-THROUGHPUT CONNECTION POOLING

PC CRM serves fast API endpoints and real-time websocket activity feeds. MariaDB's thread pool architecture efficiently handles hundreds of concurrent sales agents querying pipeline filters simultaneously with minimal CPU overhead.

4. OPEN SOURCE FREEDOM & RELIABILITY

MariaDB ensures zero vendor lock-in, active community security patches, and full compatibility with enterprise clustering solutions (Galera Cluster) for multi-master high-availability deployments.

Database Engine Verification in PC CRM

In PC CRM, all application tables are created with `ENGINE = InnoDB` and character set `utf8mb4` with collation `utf8mb4_unicode_ci`, ensuring flawless multi-language support (emojis, international contact names, and multi-currency symbols).

2. Storage Engine & Database Configuration

PC CRM enforces a dedicated MariaDB configuration profile (`/etc/mysql/mariadb.conf.d/99-pc-crm.cnf`) tailored for transaction-heavy CRM operations:

```
[mysqld]
# Full 4-byte UTF-8 support for names, emojis & international symbols
character-set-client-handshake = FALSE
character-set-server           = utf8mb4
collation-server               = utf8mb4_unicode_ci

# Transaction isolation & InnoDB memory configuration
transaction-isolation          = READ-COMMITTED
innodb_buffer_pool_size        = 1G              # Adjust to 70% of available RAM in production
innodb_file_per_table          = 1
innodb_flush_log_at_trx_commit = 2
innodb_log_buffer_size         = 64M

[mysql]
default-character-set          = utf8mb4
```

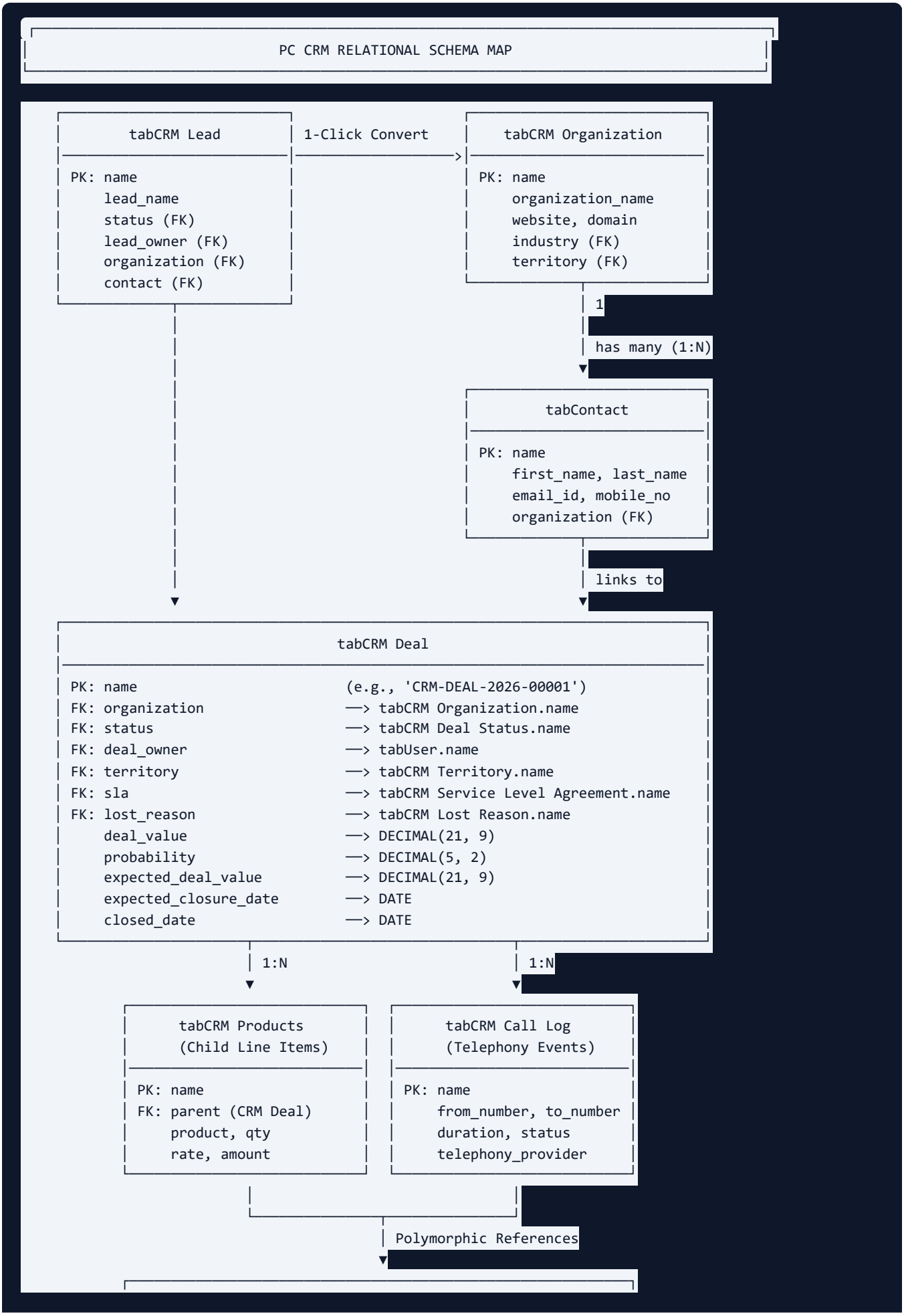
3. The Standard Document Record Layout

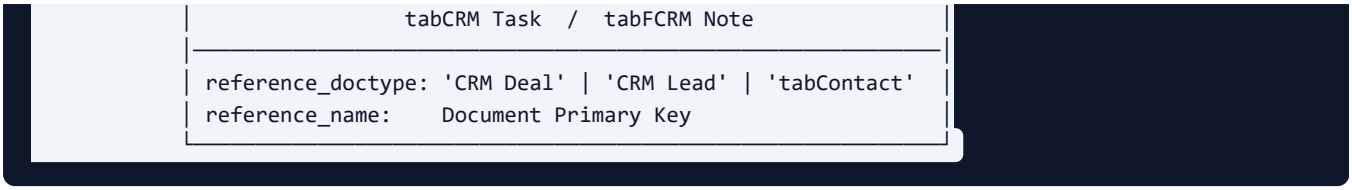
In PC CRM, every physical entity table is prefixed with **`tab`** (e.g., `tabCRM Deal` , `tabCRM Lead`). Every single table automatically inherits **7 mandatory system columns** that power document identity, ownership, audit trails, and transactional state:

Column Name	Data Type	Constraints	Description & Architectural Purpose
`name`	`VARCHAR(140)`	PRIMARY KEY	The unique business ID of the document (e.g., `CRM-DEAL-2026-00001` , `LEAD-00214` , or sanitized slug).
`creation`	`DATETIME(6)`	NOT NULL	Microsecond-accurate timestamp recording when the row was first inserted into the database.
`modified`	`DATETIME(6)`	NOT NULL, INDEXED	Timestamp of the latest update. Used for optimistic locking and cache invalidation.
`modified_by`	`VARCHAR(140)`	NOT NULL	User ID (email) who performed the most recent update.
`owner`	`VARCHAR(140)`	NOT NULL, INDEXED	User ID (email) who originally created the record. Drives user-level permission boundaries.
`docstatus`	`INT(1)`	DEFAULT 0	Document lifecycle state: `0 = Saved/Draft` , `1 = Submitted (Locked)` , `2 = Cancelled` .
`idx`	`INT(8)`	DEFAULT 0	Sequence position. Critical in child tables for preserving line-item display order.

4. Entity Relationship Model (ERD)

The relational architecture of PC CRM organizes the entire commercial lifecycle into cohesive domains: Inbound Prospecting, Active Pipeline, Account Mapping, and Execution Activities.


PC CRM RELATIONAL SCHEMA MAP
tabCRM Lead
PK: name
lead_name
status (FK)
lead_owner (FK)
organization (FK)
contact (FK)
1-Click Convert
tabCRM Organization
PK: name
organization_name
website, domain
industry (FK)
territory (FK)
1
has many (1:N)
tabContact
PK: name
first_name, last_name
email_id, mobile_no
organization (FK)
links to
tabCRM Deal
PK: name (e.g., 'CRM-DEAL-2026-00001')
FK: organization ──> tabCRM Organization.name
FK: status ──> tabCRM Deal Status.name
FK: deal_owner ──> tabUser.name
FK: territory ──> tabCRM Territory.name
FK: sla ──> tabCRM Service Level Agreement.name
FK: lost_reason ──> tabCRM Lost Reason.name
deal_value ──> DECIMAL(21, 9)
probability ──> DECIMAL(5, 2)
expected_deal_value ──> DECIMAL(21, 9)
expected_closure_date ──> DATE
closed_date ──> DATE
1:N
1:N
tabCRM Products
(Child Line Items)
PK: name
FK: parent (CRM Deal)
product, qty
rate, amount
tabCRM Call Log
(Telephony Events)
PK: name
from_number, to_number
duration, status
telephony_provider
Polymorphic References


tabCRM Task / tabFCRM Note
reference_doctype: 'CRM Deal' | 'CRM Lead' | 'tabContact'
reference_name: Document Primary Key

5. Core Sales Pipeline Tables

Table: `tabCRM Deal` (Active Revenue Opportunities)

The central transactional table storing all pipeline opportunities:

Column Name	SQL Data Type	Relationship	Field Description & Usage
`name`	`VARCHAR(140)`	PRIMARY KEY	Formatted deal number (e.g. `CRM-DEAL-.YYYY.-.#####`).
`organization`	`VARCHAR(140)`	FK → `tabCRM Organization`	The buyer company record.
`status`	`VARCHAR(140)`	FK → `tabCRM Deal Status`	Current sales pipeline stage (e.g., Prospecting, Proposal, Won, Lost).
`deal_owner`	`VARCHAR(140)`	FK → `tabUser`	The assigned account executive.
`deal_value`	`DECIMAL(21, 9)`	Value	Gross deal amount in local or quote currency.
`currency`	`VARCHAR(140)`	Currency Link	ISO Currency Code (USD, EUR, GBP, INR).
`probability`	`DECIMAL(5, 2)`	Percentage	Win probability percentage (0.00% to 100.00%).
`expected_deal_value`	`DECIMAL(21, 9)`	Computed	Forecast revenue: `deal_value * (probability / 100)`.
`expected_closure_date`	`DATE`	Target Date	Forecast target date for closing contract.
`closed_date`	`DATE`	Actual Date	Actual timestamp when marked Won or Lost.
`lost_reason`	`VARCHAR(140)`	FK → `tabCRM Lost Reason`	Mandatory reason if status is changed to Lost.
`sla`	`VARCHAR(140)`	FK → `tabCRM SLA`	Assigned Service Level Agreement ruleset.

6. Account & Stakeholder Tables

TABLE: `TABCRM ORGANIZATION`

- `name` : Primary key (Company Name / Clean ID).
- `organization_name` : `VARCHAR(140)` .
- `website` : `VARCHAR(140)` .
- `domain` : `VARCHAR(140)` (Indexed for enrichment).
- `annual_revenue` : `DECIMAL(21, 9)` .
- `no_of_employees` : `INT(11)` .
- `industry` : FK → `tabCRM Industry` .
- `territory` : FK → `tabCRM Territory` .

TABLE: `TABCONTACT`

- `name` : Primary key (Contact UID).
- `first_name` , `last_name` : `VARCHAR(140)` .
- `email_id` : `VARCHAR(140)` (Indexed for email sync).
- `mobile_no` : `VARCHAR(140)` (Indexed for phone/WhatsApp).
- `company_name` : Denormalized company string.
- `designation` : `VARCHAR(140)` (Job role).

7. Child Tables & Line Item Composition

In PC CRM, master-detail relationships (such as product line items on a deal) use **Child Tables**. Child tables do not exist independently; they are bound to a parent document via three structural columns:

Child Column	Type	Architectural Function
`parent`	`VARCHAR(140)`	Foreign Key pointing to the parent document's `name` (e.g., `CRM-DEAL-2026-00001`).
`parenttype`	`VARCHAR(140)`	The DocType of the parent (e.g., `CRM Deal`).
`parentfield`	`VARCHAR(140)`	The specific fieldname in the parent document (e.g., `products`).

Schema: `tabCRM Products` (Deal Line Items)

```
CREATE TABLE `tabCRM Products` (
  `name` VARCHAR(140) NOT NULL PRIMARY KEY,
  `creation` DATETIME(6) NOT NULL,
  `modified` DATETIME(6) NOT NULL,
  `modified_by` VARCHAR(140) NOT NULL,
  `owner` VARCHAR(140) NOT NULL,
  `docstatus` INT(1) NOT NULL DEFAULT 0,
  `parent` VARCHAR(140) NOT NULL,
  `parenttype` VARCHAR(140) NOT NULL,
  `parentfield` VARCHAR(140) NOT NULL,
  `idx` INT(8) NOT NULL DEFAULT 0,
  `product` VARCHAR(140) DEFAULT NULL,
  `product_name` VARCHAR(140) DEFAULT NULL,
  `qty` DECIMAL(21, 9) NOT NULL DEFAULT 1.000000000,
  `rate` DECIMAL(21, 9) NOT NULL DEFAULT 0.000000000,
  `discount_percentage` DECIMAL(5, 2) NOT NULL DEFAULT 0.00,
  `amount` DECIMAL(21, 9) NOT NULL DEFAULT 0.000000000,
  INDEX `parent` (`parent`),
  INDEX `product` (`product`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

8. Activity, Task & Omnichannel Storage

Activities are linked polymorphically to any entity in PC CRM:

- **`tabCRM Task`:** Stores actionable to-dos with `due_date`, `priority` (Low, Medium, High, Urgent), `assigned_to`, and polymorphic links: `reference_doctype` and `reference_name`.
- **`tabFCRM Note`:** Stores rich-text notes in a `LONGTEXT` column named `content`, with a `pinned` (0 or 1) boolean flag for sticky intelligence.
- **`tabCRM Call Log`:** Stores telephony logs (Twilio/Exotel) with `from_number`, `to_number`, `duration`, `call_type`, `status`, and `recording_url`.

9. Service Level Agreements (SLA) Schema

PC CRM enforces response and resolution deadlines using a multi-tiered relational SLA model:

MASTER: `tabCRM Service Level Agreement`

- `name`: Unique SLA policy name (e.g., Enterprise 24/7).
- `default_priority`: Standard priority (Medium).
- `holiday_list`: FK → `tabCRM Holiday List`.
- `service_days`: Child table of working business hours.

CHILD: `tabCRM Service Level Priority`

- `parent`: Links to master SLA.
- `priority`: Urgent, High, Medium, Low.
- `response_time`: Target first touch seconds.
- `resolution_time`: Target resolution seconds.

10. Territory & Sales Hierarchy Data Structures

PC CRM utilizes nested trees and reporting hierarchies:

- **`tabCRM Territory`:** Stored as a Modified Preorder Tree Traversal (Nested Set) using `lft` and `rgt` integer columns, enabling instantaneous subtree queries (e.g., retrieving all deals in North America and all child states in a single indexed query).
- **`tabCRM Sales Hierarchy`:** Directed adjacency tree mapping sales reps to team leads and regional managers. Fuels manager rollup visibility without recursive join bottlenecks.

11. The Single Table Pattern (`tabSingles`)

System settings documents (such as `FCRM Settings` or `PC CRM Global Settings`) do not require full relational tables. Instead, PC CRM stores them in a high-efficiency Key-Value table called **`tabSingles`**:

Column	Type	Description & Example Data
`doctype`	`VARCHAR(140)`	The settings configuration name (e.g. `FCRM Settings`).
`field`	`VARCHAR(140)`	The setting key (e.g. `enable_forecasting`, `currency`).
`value`	`LONGTEXT`	The stored setting value (e.g. `1`, `USD`, `{"theme":"dark"}`).

12. Audit Trails, Versions & Change Tracking

PC CRM maintains complete compliance accountability through dedicated audit tables:

- **`tabVersion`:** Captures a full JSON delta whenever any document is saved. It records `ref_doctype`, `docname`, `data` (showing old vs. new values), `owner`, and timestamp. This allows 1-click historical rollbacks.
- **`tabActivity Log`:** Logs user sessions, login timestamps, IP addresses, data export events, and bulk actions.

13. Indexing Strategies & Query Optimization

For high-scale databases handling millions of leads and deals, PC CRM implements rigorous indexing policies:

Index Type	Columns Indexed	Query Optimization Benefit
Clustered Primary Key	`PRIMARY KEY (`name`)`	Instant $O(1)$ single-record retrieval for API and detail views.
Status & Owner Composite	`INDEX (`status`, `deal_owner`)`	Accelerates Kanban pipeline filtering by stage and representative.
Parent FK Indexes	`INDEX (`parent`, `parenttype`)`	Ensures instant joins when rendering line items on deals and quotes.
Search Text Indexes	`INDEX (`email_id`), INDEX (`mobile_no`)`	Sub-millisecond lookup during inbound telephony and email matching.

14. Database Maintenance, Backups & Recovery

PC CRM provides automated MariaDB snapshot utilities:

HOT DATABASE BACKUP COMMAND

Creates a full gzip snapshot with transactional consistency:

```
# Snapshot database + files
bench --site crm.localhost backup --with-files
```

Generates an SQL dump in `sites/crm.localhost/private/backups/`.

DATABASE RESTORE COMMAND

Restores database from an existing SQL backup file:

```
# Restore into site
bench --site crm.localhost restore \
  /path/to/backup.sql.gz
```